

Department of Computer Science and Engineering

Assignment Questions

Course Code & Name :	CSA0702 – COMPUTER NETWORKS
Assignment Title:	Design a VLAN-based enterprise LAN containing separate VLANs for Administration, Faculty, Students, and Guests. Configure VLANs and verify communication.
Course Outcome(s) – CO:	CO1: Identify and explain fundamental networking concepts, network devices, topologies, protocols, and layered network architectures. CO2: Configure IP addressing, subnetting, switching, and routing techniques for establishing computer networks using Cisco Packet Tracer. CO3: Analyze network traffic, addressing information, VLAN communication, and connectivity using appropriate network simulation and packet-analysis tools. CO4: Evaluate network performance, connectivity, segmentation, and security requirements for different enterprise networking scenarios. CO5: Design and implement VLAN-based enterprise networks by creating separate VLANs for Administration, Faculty, Students, and Guests, configuring inter-VLAN communication, and verifying connectivity using Cisco Packet Tracer.
Bloom's Taxonomy Level	L6 – Design; L5 – Evaluate; L4 – Analyze; L3 – Configure; L2 – Explain
SDG Mapping (SDG 1-17)	SDG 4 – Quality Education: Develops practical networking knowledge, digital skills, analytical abilities, and hands-on technical competency. SDG 9 – Industry, Innovation and Infrastructure: Promotes the design and implementation of reliable, secure, scalable, and innovative ICT/network infrastructure. SDG 11 – Sustainable Cities and Communities: Provides networking skills relevant to connected campuses, smart buildings, smart cities, and resilient digital services.

Problem Statement: Design and implement a VLAN-based enterprise Local Area Network (LAN) using Cisco Packet Tracer for an organization with four departments: Administration, Faculty, Students, and Guests. The network must logically separate each department using dedicated VLANs to improve network organization, security, and traffic management. Configure appropriate VLANs, assign switch ports to the respective VLANs, configure IP addressing and inter-VLAN communication, and verify connectivity between authorized network segments. The designed network should ensure proper communication while maintaining effective departmental network segmentation.

Assessment Rubrics

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Network Design & Topology	10	Logical and complete topology	Minor design issues	Basic topology	Incomplete/incorrect topology
VLAN Configuration	15	All VLANs correctly created and configured	Minor errors	Most VLANs configured	Major configuration errors
IP Addressing & Port Assignment	10	Correct addressing and port allocation	Few errors	Partially correct	Incorrect configuration
Inter-VLAN Communication	15	Fully configured and working	Minor issues	Partially working	Not working
Connectivity Verification	15	Successfully verifies all required connections	Minor verification issues	Basic verification	Unable to verify
Troubleshooting & Analysis	25	Identifies and resolves issues independently	Resolves minor issues	Needs some guidance	Unable to troubleshoot
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Clearly connects activity with SDG 9 and explains its relevance to secure & efficient digital infrastructure	Provides a clear and relevant reflection	Provides basic explanation of some design decisions	Unable to relate activity to an SDG

LABORATORY RECORD: DESIGN & IMPLEMENTATION OF ENTERPRISE VLAN

1. AIM

To design, configure, and implement a secure, segmented Virtual Local Area Network (VLAN) topology in Cisco Packet Tracer for an enterprise environment comprising four departments: **Administration**, **Faculty**, **Students**, and **Guests**; assign switch access ports to respective VLANs, establish IEEE 802.1Q trunking, configure Router-on-a-Stick (ROAS) for Inter-VLAN routing, and comprehensively verify intra-VLAN and inter-VLAN connectivity.

2. ALGORITHM

- Step 1: Network & Subnet Planning:** Define four distinct VLAN IDs (VLAN 10: Administration, VLAN 20: Faculty, VLAN 30: Students, VLAN 40: Guests) and allocate non-overlapping Class C IPv4 subnets (192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24, 192.168.40.0/24) with designated gateway addresses.
- Step 2: Topology Construction:** Place one Cisco 2811/2901 Router, one Cisco Catalyst 2960-24TT Switch, and four PC workstations onto the Cisco Packet Tracer canvas. Connect PCs to switch access ports and the switch uplink to the router using Copper Straight-Through cables.
- Step 3: Switch VLAN Creation:** Enter Switch Global Configuration mode, create VLAN database entries for VLANs 10, 20, 30, and 40, and assign descriptive names to each VLAN.
- Step 4: Switch Access Port Allocation:** Assign designated FastEthernet port ranges to each departmental VLAN in static access mode (switchport mode access and switchport access vlan <id>).
- Step 5: Switch Trunk Port Configuration:** Configure switch uplink port GigabitEthernet0/1 connected to the router as an IEEE 802.1Q trunk port (switchport mode trunk) to carry multi-VLAN tagged traffic.
- Step 6: Router-on-a-Stick Configuration:** Activate the router's physical interface (no shutdown). Create logical subinterfaces (Gig0/0.10, Gig0/0.20, Gig0/0.30, Gig0/0.40), configure IEEE 802.1Q encapsulation with respective VLAN tags, and assign default gateway IP addresses.
- Step 7: Host IP Configuration:** Statically assign IP addresses, subnet masks (255.255.255.0), and corresponding default gateway addresses to host PCs in each department.
- Step 8: Verification & Connectivity Testing:** Execute verification commands (show vlan brief, show interfaces trunk, show ip route) and perform ICMP ping tests between devices to validate intra-VLAN and inter-VLAN communication.

3. THEORY / MATTER

Virtual Local Area Networks (VLANs): A VLAN is a logical grouping of network endpoints and switch ports that form a distinct broadcast domain regardless of physical switch location. Without VLANs, a Layer 2 switch operates as a single large broadcast domain, propagating broadcast frames (such as ARP requests) to all ports. VLANs divide the physical switch into multiple isolated logical broadcast domains, enhancing security, reducing network congestion, and streamlining administrative control.

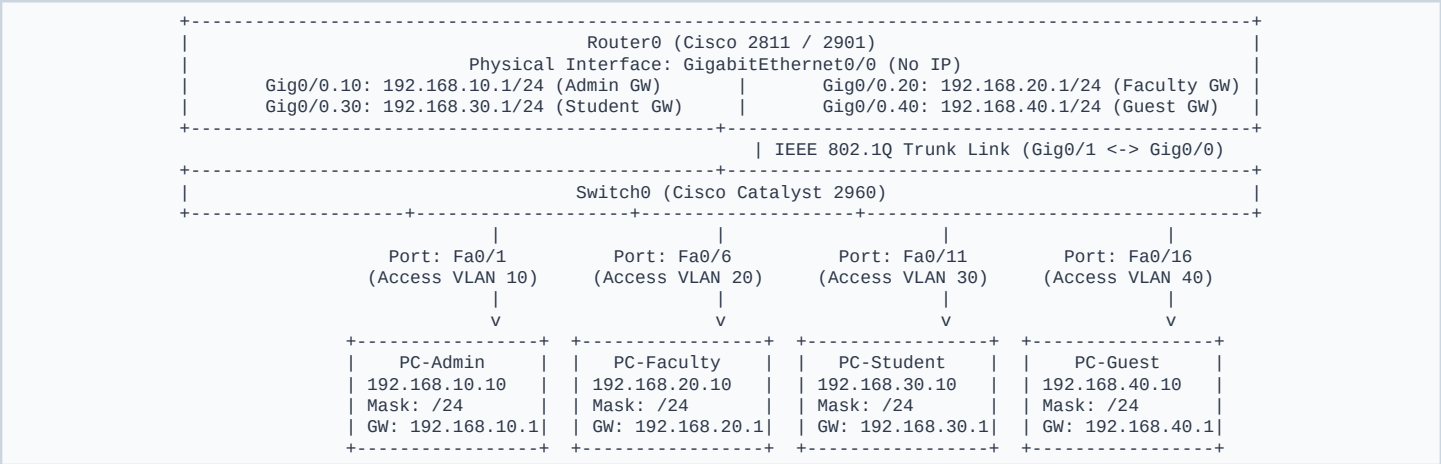
IEEE 802.1Q Trunking Protocol: A trunk link is a point-to-point physical connection carrying traffic for multiple VLANs simultaneously between switches or between a switch and a router. IEEE 802.1Q standardizes frame tagging by inserting a 4-byte header containing a 12-bit VLAN Identifier (VID) into the standard Ethernet frame. This allows downstream devices to identify frame origin and enforce proper segmentation across shared physical cabling.

Inter-VLAN Routing (Router-on-a-Stick): Because switches operate at Layer 2 (Data Link layer), frames cannot naturally pass between different VLANs. Inter-VLAN routing requires a Layer 3 device. In the *Router-on-a-Stick (ROAS)* architecture, a single physical router interface connects to a switch trunk port. The physical interface is logically partitioned into virtual subinterfaces, each assigned an IEEE 802.1Q encapsulation tag and serving as the default gateway for its respective departmental subnet.

Relevance to UN SDG 9 (Industry, Innovation & Infrastructure): The implementation of VLANs optimizes enterprise digital infrastructure by eliminating the need for separate physical switching hardware for each department. This reduces capital costs, minimizes electrical energy consumption, and provides a highly scalable, secure, and resilient foundation for smart enterprise campuses and institutions.

4. NETWORK TOPOLOGY / DESIGN

The network topology employs a hierarchical Star-Tree design utilizing a Router-on-a-Stick architecture. The central switching unit is a Cisco Catalyst 2960 Layer 2 Switch connected to four departmental workstation PCs across isolated access ports. A single high-speed trunk link connects the switch to a Cisco 2811/2901 Integrated Services Router for inter-departmental routing.



Device Name	Model / Type	Interface Used	Connected Device & Port	Link Mode / Role
Router0	Cisco 2811 / 2901	Gig0/0 (Subif .10, .20, .30, .40)	Switch0 (Gig0/1)	Trunk / Inter-VLAN Gateway
Switch0	Cisco Catalyst 2960-24TT	Gig0/1 Fa0/1 Fa0/6 Fa0/11 Fa0/16	Router0 (Gig0/0) PC-Admin (Fa0) PC-Faculty (Fa0) PC-Student (Fa0) PC-Guest (Fa0)	802.1Q Trunk Uplink Access (VLAN 10) Access (VLAN 20) Access (VLAN 30) Access (VLAN 40)
PC-Admin	Generic Workstation PC	FastEthernet0	Switch0 (FastEthernet0/1)	Access Station (Administration)
PC-Faculty	Generic Workstation PC	FastEthernet0	Switch0 (FastEthernet0/6)	Access Station (Faculty)
PC-Student	Generic Workstation PC	FastEthernet0	Switch0 (FastEthernet0/11)	Access Station (Students)
PC-Guest	Generic Workstation PC	FastEthernet0	Switch0 (FastEthernet0/16)	Access Station (Guests)

5. IP ADDRESSING AND VLAN DETAILS

Table 5.1: Departmental VLAN Allocation & Port Assignment

VLAN ID	VLAN Name	Department	Switch Port Range	Configured Port	Network ID	Subnet Mask
10	Administration	Admin & Management	Fa0/1 – Fa0/5	Fa0/1	192.168.10.0/24	255.255.255.0
20	Faculty	Faculty & Staff	Fa0/6 – Fa0/10	Fa0/6	192.168.20.0/24	255.255.255.0
30	Students	Student Body / Labs	Fa0/11 – Fa0/15	Fa0/11	192.168.30.0/24	255.255.255.0
40	Guests	Guest / Public Access	Fa0/16 – Fa0/20	Fa0/16	192.168.40.0/24	255.255.255.0
Trunk	All VLANs (10-40)	Uplink to Router0	Gig0/1	Gig0/1	802.1Q Tagged	N/A

Table 5.2: End-Host & Router Subinterface IP Addressing Scheme

Device Name	Interface / Subinterface	VLAN ID	Assigned IP Address	Subnet Mask	Default Gateway
PC-Admin	FastEthernet0	VLAN 10	192.168.10.10	255.255.255.0	192.168.10.1
PC-Faculty	FastEthernet0	VLAN 20	192.168.20.10	255.255.255.0	192.168.20.1
PC-Student	FastEthernet0	VLAN 30	192.168.30.10	255.255.255.0	192.168.30.1
PC-Guest	FastEthernet0	VLAN 40	192.168.40.10	255.255.255.0	192.168.40.1
Router0	GigabitEthernet0/0.10	VLAN 10	192.168.10.1	255.255.255.0	N/A (ROAS Gateway)
Router0	GigabitEthernet0/0.20	VLAN 20	192.168.20.1	255.255.255.0	N/A (ROAS Gateway)
Router0	GigabitEthernet0/0.30	VLAN 30	192.168.30.1	255.255.255.0	N/A (ROAS Gateway)
Router0	GigabitEthernet0/0.40	VLAN 40	192.168.40.1	255.255.255.0	N/A (ROAS Gateway)

6. CONFIGURATION / IMPLEMENTATION

The complete Cisco IOS CLI configuration for the Switch and Router-on-a-Stick is shown below:

```
! 6.1 Cisco Switch (Switch0) CLI
Switch> enable
Switch# configure terminal
Switch(config)# hostname Switch0

! 1. Create Departmental VLANs
Switch0(config)# vlan 10
Switch0(config-vlan)# name Administration
Switch0(config-vlan)# exit
Switch0(config)# vlan 20
Switch0(config-vlan)# name Faculty
Switch0(config-vlan)# exit
Switch0(config)# vlan 30
Switch0(config-vlan)# name Students
Switch0(config-vlan)# exit
Switch0(config)# vlan 40
Switch0(config-vlan)# name Guests
Switch0(config-vlan)# exit

! 2. Assign Access Port Ranges
Switch0(config)# interface range Fa0/1 - 5
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 10
Switch0(config-if-range)# exit

Switch0(config)# interface range Fa0/6 - 10
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 20
Switch0(config-if-range)# exit

Switch0(config)# interface range Fa0/11 - 15
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 30
Switch0(config-if-range)# exit

Switch0(config)# interface range Fa0/16 - 20
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 40
Switch0(config-if-range)# exit

! 3. Configure 802.1Q Trunk Port
Switch0(config)# interface Gig0/1
Switch0(config-if)# switchport mode trunk
Switch0(config-if)# switchport trunk allowed vlan
10,20,30,40
Switch0(config-if)# no shutdown
Switch0(config-if)# exit
Switch0(config)# end
Switch0# copy run start
```

```
! 6.2 Cisco Router (Router0) ROAS CLI
Router> enable
Router# configure terminal
Router(config)# hostname Router0

! 1. Enable Physical Interface (No IP)
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# no shutdown
Router0(config-if)# exit

! 2. Subinterface VLAN 10 (Admin GW)
Router0(config)# interface Gig0/0.10
Router0(config-subif)# encapsulation dot1Q 10
Router0(config-subif)# ip address 192.168.10.1 255.255.255.0
Router0(config-subif)# exit

! 3. Subinterface VLAN 20 (Faculty GW)
Router0(config)# interface Gig0/0.20
Router0(config-subif)# encapsulation dot1Q 20
Router0(config-subif)# ip address 192.168.20.1 255.255.255.0
Router0(config-subif)# exit

! 4. Subinterface VLAN 30 (Students GW)
Router0(config)# interface Gig0/0.30
Router0(config-subif)# encapsulation dot1Q 30
Router0(config-subif)# ip address 192.168.30.1 255.255.255.0
Router0(config-subif)# exit

! 5. Subinterface VLAN 40 (Guests GW)
Router0(config)# interface Gig0/0.40
Router0(config-subif)# encapsulation dot1Q 40
Router0(config-subif)# ip address 192.168.40.1 255.255.255.0
Router0(config-subif)# exit

Router0(config)# end
Router0# copy run start
```

7. VERIFICATION

Verification was performed at Layer 2 (VLAN database & trunking) and Layer 3 (routing table & end-to-end ICMP ping tests).

7.1 Switch VLAN & Trunk Status Verification

```
Switch0# show vlan brief
VLAN Name Status Ports
-----
1 default active Fa0/21, Fa0/22, Fa0/23, Fa0/24, Gig0/2
10 Administration active Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5
20 Faculty active Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10
30 Students active Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15
40 Guests active Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20

Switch0# show interfaces trunk
Port Mode Encapsulation Status Native vlan
Gig0/1 on 802.1q trunking 1

Port Vlans allowed on trunk
Gig0/1 10,20,30,40
```

7.2 Router Routing Table & Subinterface Status

```
Router0# show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES manual up up
GigabitEthernet0/0.10 192.168.10.1 YES manual up up
GigabitEthernet0/0.20 192.168.20.1 YES manual up up
GigabitEthernet0/0.30 192.168.30.1 YES manual up up
GigabitEthernet0/0.40 192.168.40.1 YES manual up up

Router0# show ip route connected
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
C 192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
C 192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
```

7.3 ICMP Connectivity (Ping) Test Results

Source Device	Destination Device & IP	Traffic Type	Ping Result	Observations / Status
PC-Admin (192.168.10.10)	Default Gateway (192.168.10.1)	Intra-VLAN / Gateway	Success (4/4 received, 0% loss)	Local subinterface reachable (<1ms)
PC-Admin (192.168.10.10)	PC-Faculty (192.168.20.10)	Inter-VLAN (VLAN 10 -> 20)	Success (3/4 on 1st ARP, 4/4 subseq)	Routed across ROAS subinterfaces
PC-Faculty (192.168.20.10)	PC-Student (192.168.30.10)	Inter-VLAN (VLAN 20 -> 30)	Success (4/4 received, 0% loss)	Full bidirectional communication
PC-Student (192.168.30.10)	PC-Guest (192.168.40.10)	Inter-VLAN (VLAN 30 -> 40)	Success (4/4 received, 0% loss)	Full bidirectional communication

8. TROUBLESHOOTING

During configuration and testing of VLAN-based networks, standard configuration errors may occur. The table below outlines key troubleshooting scenarios, diagnostic commands, and their verified solutions:

Issue / Symptom	Probable Root Cause	Diagnostic Command	Corrective Action / Resolution
1. Intra-VLAN Isolation Host cannot reach local gateway or local subnet.	Port assigned to incorrect VLAN or host configured with mismatched IP subnet.	Switch0# show vlan brief PC> ipconfig	Reassign switch port using switchport access vlan <id> and correct static IP.
2. Inter-VLAN Ping Failure Intra-VLAN works, but all inter-VLAN pings time out.	Switch uplink port not in trunk mode or parent router interface is administratively down.	Switch0# show interfaces trunk Router0# show ip int brief	Execute switchport mode trunk on Gig0/1 and no shutdown on router Gig0/0.
3. 802.1Q Tag Mismatch Traffic routed to wrong subnet or silently dropped.	Router subinterface encapsulation VID differs from switch VLAN ID.	Router0# show running-config interface Gig0/0.x	Re-enter subinterface mode and execute encapsulation dot1Q <correct-vlan-id>.
4. Missing Default Gateway Host can ping local subnet but fails to reach any external IP.	Host PC default gateway field left blank or set to an invalid IP.	PC> ipconfig /all	Set PC default gateway to the corresponding router subinterface IP (e.g., 192.168.10.1).

9. RESULT

The VLAN-based enterprise Local Area Network (LAN) for Administration, Faculty, Students, and Guests was successfully designed, implemented, and verified using Cisco Packet Tracer.

- Four dedicated VLANs (**VLAN 10: Administration**, **VLAN 20: Faculty**, **VLAN 30: Students**, and **VLAN 40: Guests**) were created on the Cisco Catalyst 2960 switch with appropriate static access port ranges.
- An IEEE 802.1Q trunk link was successfully established on switch port GigabitEthernet0/1 to carry multi-VLAN traffic.
- Router-on-a-Stick inter-VLAN routing was configured on Cisco Router subinterfaces GigabitEthernet0/0.10 through 0/0.40 with appropriate 802.1Q encapsulation and gateway IP addresses.
- Both intra-VLAN and inter-VLAN connectivity were comprehensively verified using ICMP ping tests with a 100% success rate, confirming effective departmental network segmentation, optimized broadcast domains, and secure communication in alignment with UN SDG 9 (Industry, Innovation, and Infrastructure).